

東京 2025 LINUX PLUMBERS CONFERENCE

TOKYO, JAPAN / DECEMBER 11-13, 2025

32bit and noMMU Linux BoF

Still alive and running in the background.

But people use devices with it every day… without knowing it



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Hello all !

"Ruinland" ChuanTzu Tsai (Andes Technology, RISC-V 32)

D Jeff Dionne (Coresemi, SuperH, J-Core)

Geert Uytterhoeven (m68k)

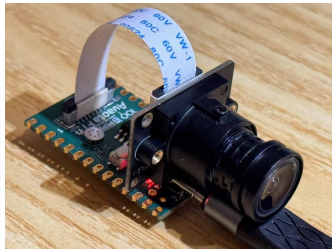


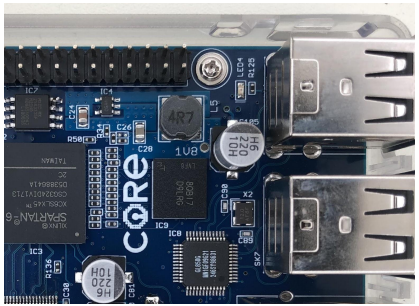
32 is the new 8-bit

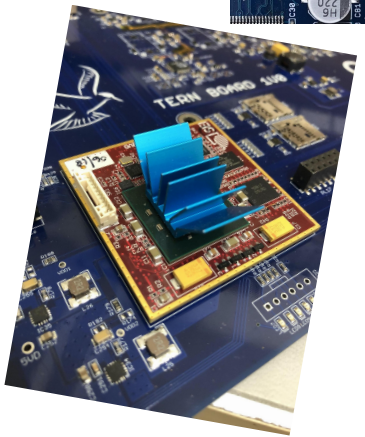
RISC-V and SuperH are still very well adopted and thriving.

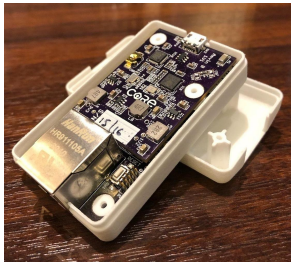
Nanny-cams, automotive controller, low-end internet connecting device ..... a lot of them are running Linux for their tasks.











“Userspace aliveness”

Names carrying weight are pointing fingers on userspace SW ecosystem for coping with 2038 issues - -

- OpenWRT is still well alive and maintaining the healthiness of programs.
- buildroot housed noMMU Linux “distributions.”
- I personally ported Alpine Linux to RV32 and is willing to upgrade it.
- musl libc is of course maintained specifically for such use cases, most things are built on that today

The unmaintained projects simply aren’t used by 32bit adopters anymore.

CAN bus driver for STM32 noMMU Linux

FOSDEM 2024 “Linux CAN upstreaming on MMU-less systems” :

https://github.com/torvalds/linux/commit/f00647d8127be4d3f37f7e07dace24c04689ec63

stm32f469-disco



Modify Buildroot configuration
Create Linux driver - **dual CAN**

stm32f769-disco



Create Buildroot configuration
Modify Linux driver - **single CAN**

rv64ilp32

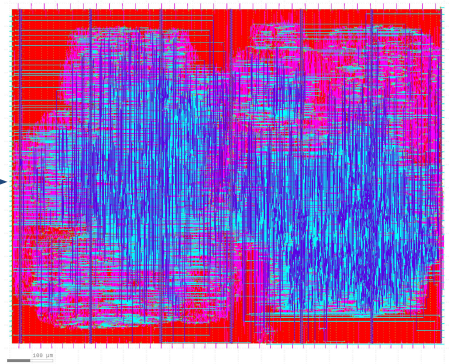
To run 32bit userspace on 64bit hardware :

https://lpc.events/event/17/contributions/1475/attachments/1186/2442/rv64ilp32_%20Run%20ILP32%20on%20RV64%20ISA.pdf

https://fedoraproject.org/wiki/Architectures/RISC-V/64ILP32

Small memory footprint is crucial for embedded Linux scenarios.

J32+FPU
130nm



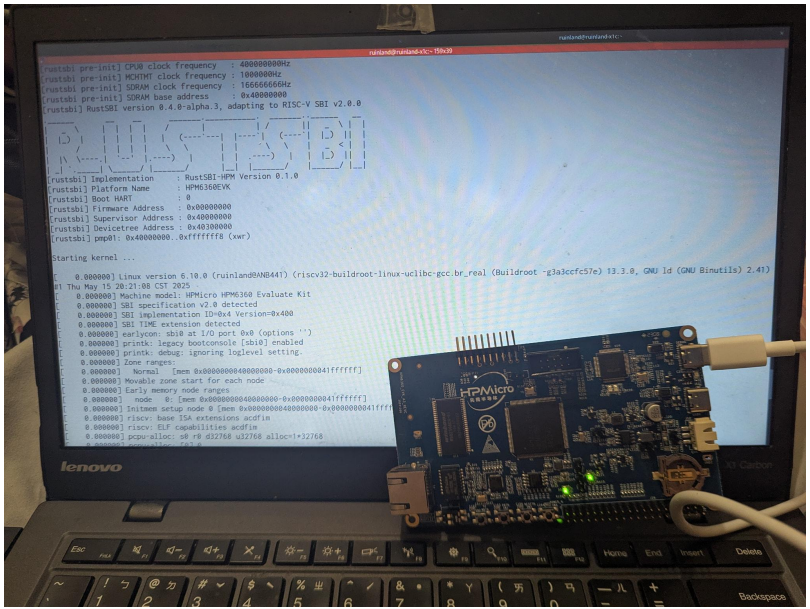
- Memory footprint and bandwidth are a major driver for hardware designs.
- Don't forget the process node the chips are fabricated in is important too
  - An application specific Linux capable RV32 or J-Core design can be fabricated for $15,000
  - All open hardware, prototyped on your desk. Your chip, your hardware, your security.

noMMU Linux on newly manufactured RV32 hardware

Automotive platforms are very keen to small but power efficient controllers.

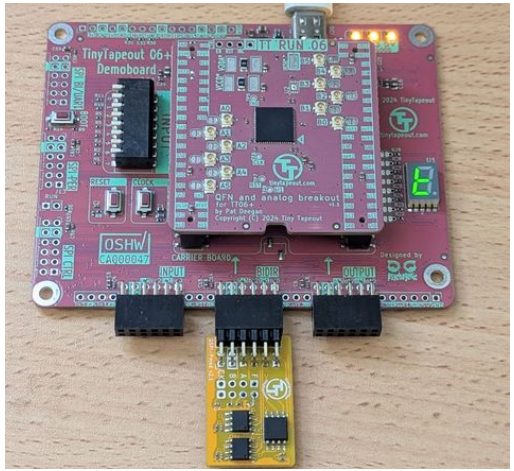
Compare to noMMU Linux, RTOSes requires a tons of work to effort to port complicated programs.

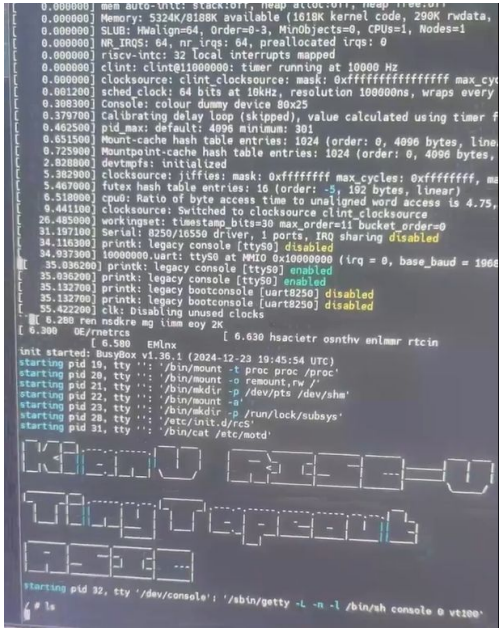
buildroot src



Educational purpose

TinyTapeout kianV SoC : end-to-end open source-ly manufactured IC, running 32bit noMMU mainline Linux (with minimal driver modifications).





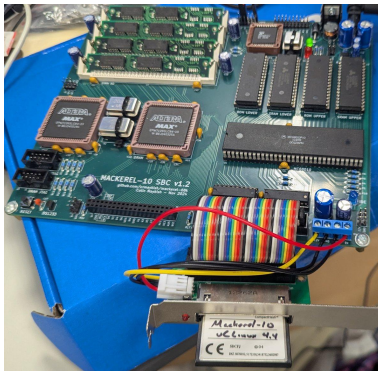
Don't forget hobbyist works. This is where innovation comes from

Mackerel boards, made by maykish, the M68Ks are still booting Linux :

https://github.com/crmaykish/mackerel-68k

And he even has gotten a basic port of the latest Linux kernel (v6.17) running on Mackerel-30. It boots to userspace and has a Toybox-based init and interactive shell !

https://hackaday.io/project/183861-mackerel-68k-linux-sbcs/log/244041-linux-617-on-mackerel-30





Current issues for maintenance

Real Hardware for testing :

- SuperH J-Core in FPGA and on multiple silicon processes, in multiple hardware is available.
- m68k are still well-stocked and there are other new hardardware, such as Apollo 68080.
- RISC-V has plenty of hardware manufactured.

Workforce perspective :

- Hardware vendors are tend to maintain only a few releases of Linux.
    - It's important for the community to pay attention to what embedded engineers actually use
- SLTS from CIP is what many vendors are looking for.
- Regular clean-up and testing are needed.

32 is the new 8-bit

RISC-V and SuperH are still very well adopted and thriving.

Cams, automotive controller, energy monitors, critical infrastructure, real time devices, other IIoT devices .....

A lot of them are running Linux for their tasks.

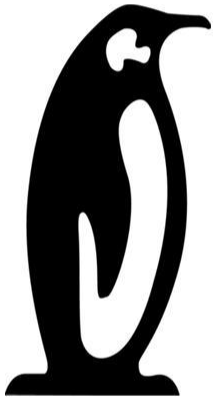
But it's important to understand that the reason is it is easy to build things with this technology

You can get OpenHardware/Gateware versions of all of this, that are supported in mainline Linux

Development boards and kits are available

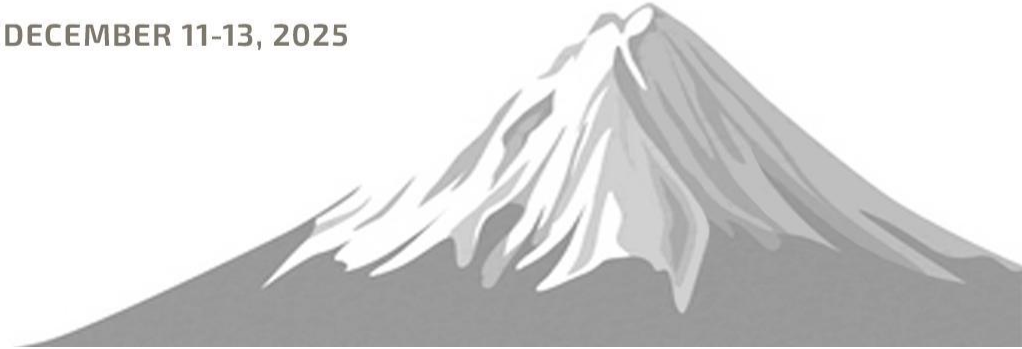
Yes, there is commercial support... but the community drives the innovation

To suggest that 32bit and nommu is irrelevant today is just incorrect.



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Backup

Place holders

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